Statistics & Econometrics 2025

Tutorial 1: Problem Set

Emanuel Moench* Robin Schaal†

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*Professor of Financial- and Monetary Economics, Frankfurt School of Finance & Management, Centre for Central Banking, CEPR, E-Mail: e.moench@fs.de

†PhD Candidate in Finance, Frankfurt School of Finance & Management, Centre for Central Banking, E-Mail: r.schaal@fs.de

1 Ordinary Least Squares Regression

Exercises to deepen understanding of OLS estimator for univariate linear regression and corresponding standard errors.

a) You have 12 years of time-series data on your firm's portfolio and want to compare its performance to a benchmark portfolio. Using these data, your intern has computed the sample annual average of your firm's portfolio return to equal 8% and that of the benchmark portfolio to equal 6%. (S)he further calculated the sample annual standard deviation of your firm's portfolio return was 5% and that of the benchmark 8%. Is your manager right if she says the firm's portfolio performs better while similarly being safer than the benchmark portfolio?
Solution: Call the benchmark portfolio x and the firm's portfolio y. Then, compare the sample means and variances of y and x. If the sample mean $\bar{y} = \frac{1}{T}\sum_{t=1}^{T} y_t$ is greater than the sample mean $\bar{x} = \frac{1}{T}\sum_{t=1}^{T} x_t$, then your firm's portfolio has delivered higher returns in the sample. This is the case for your firm's portfolio. Likewise, if the sample variance $Var(x) = \frac{1}{T-1}\sum_{t=1}^{T}(x_t - \bar{x})^2$ is greater than the sample variance $Var(y) = \frac{1}{T-1}\sum_{t=1}^{T}(y_t - \bar{y})^2$, then your firm's portfolio has been less risky than the benchmark over the sample considered. This is the case here as well for your firm's portfolio. Therefore, your manager's claim is correct.

b) Now your manager asks you to find out what happens to the firm's portfolio if the return of the benchmark portfolio changes by 1 %. In addition, (s)he is interested if your firm's portfolio earns a larger return than the benchmark portfolio, once we adjust for exposure to the benchmark portfolio. To help you in your task, your intern calculated that the sample correlation between the two portfolios is around 80%. Calculate the β and α of your firm's portfolio against the benchmark.

Solution: To estimate the quantities your manager is interested in you run a linear regression of your firms' portfolio return on the benchmark portfolio return:

$$y_t = \alpha + \beta x_t + u_t.$$

Estimating the regression using OLS, your estimates for α and β will be equal to

$$\hat{\alpha} = \bar{y} - \hat{\beta}\bar{x}$$

and

$$\hat{\beta} = \frac{Cov(y, x)}{Var(x)}$$

where $\bar{x}$ and $\bar{y}$ denote the sample means of the two portfolios, $Var(x)$ the sample variance of the benchmark portfolio return and $Cov(y, x)$ the sample covariance between the two portfolio returns. You have been given information on the sample correlation and sample standard deviations for the two portfolio returns. So before you can use this estimator, you need to convert these into the sample variances and sample covariance. The definition of the variance is square of the standard deviation. So a sample standard deviation of 5% (8%) translates into sample variances of 25 and 64, respectively. The correlation is defined as the covariance divided by the product of the standard deviations:

$$Corr(x, y) = \frac{Cov(x, y)}{\sigma_x \sigma_y}$$

where $\sigma_x = \sqrt{Var(x)}$ and $\sigma_y = \sqrt{Var(y)}$. Hence, you can impute the sample covariance as

$$Cov(x, y) = Corr(x, y) \cdot \sigma_x \cdot \sigma_y$$

Given the numbers above, this is

$$Cov(x, y) = 0.8 \cdot 8 \cdot 5 = 32$$

Now you have all the ingredients to compute the coefficients of the linear relationship between y and x. Plugging into the OLS formulae for $\hat{\alpha}$ and $\hat{\beta}$ you obtain

$$\hat{\beta} = \frac{Cov(y, x)}{Var(x)} = \frac{32}{64} = 0.5$$

and

$$\hat{\alpha} = \bar{y} - \hat{\beta}\bar{x} = 8\% - 0.5 \cdot 6\% = 5\%$$

Hence, you tell your manager that the firm's portfolio has a beta of 0.5 and an alpha of 5%.

c) Now, use your estimated alpha and beta to predict what return your firm's portfolio would likely earn in a year when the benchmark portfolio had lost 10% and in a year when the benchmark portfolio had gained 10%.

Solution: You can use the estimated relationship between the two portfolios to answer the question what the expected return on your firm's portfolio is in a year when the benchmark portfolio had lost or gained 10%.

$$\hat{y}_t = \hat{\alpha} + \hat{\beta}x_t = 5\% + 0.5 \cdot (-10\%) = 0\%$$

$$\hat{y}_t = \hat{\alpha} + \hat{\beta}x_t = 5\% + 0.5 \cdot (10\%) = 10\%$$

So based on the historical relationship observed in the data, in a year when the benchmark portfolio had lost 10% your firm's portfolio would have neither made a gain nor a loss. In a year when the benchmark gained 10%, your firm's portfolio would also deliver 10%.

d) Your manager is surprised and says that for a correlation of 80% between the original portfolio and the benchmark portfolio, (s)he would have expected a beta equal to 0.8

How can you test your manager's hypothesis? And can you reject your manager's hypothesis at the 5% level? For this analysis, assume that the estimated standard error for β_y was $SE(\hat{\beta}_y) = 0.125$.

<u>Solution:</u> Your manager's null hypothesis is $H_0 : \hat{\beta}_y = 0.8 \quad \text{vs.} \quad H_1 : \hat{\beta}_y \neq 0.8$

You can test your manager's hypothesis with the test of significance approach that you remember from your Statistics class. Luckily, you recall that the t-test statistic is given by

$$t_\beta = \frac{\hat{\beta}_y - \beta^*}{SE(\hat{\beta}_y)} \sim t_{T-2}$$

You have twelve years of data, so $T-2 = 10$. Moreover, β^* denotes the hypothesized true value of β which in this case equals $\beta^* = 0.8$. With these inputs, your test statistic is

$$t_\beta = \frac{\hat{\beta}_y - \beta^*}{SE(\hat{\beta}_y)} = \frac{0.5 - 0.8}{0.125} = -2.4$$

You have to compare this to the critical value for a two-sided t-test with $12-2 = 10$ degrees of freedom: $t_{crit} = t_{10;5\%} = 2.228$. Hence, since $t_\beta = -2.4 < -t_{crit} = -2.228$ you can reject your manager's (null) hypothesis at the 5% level.

To make your manager a little happier you note, however, that $t_{crit} = t_{10;1\%} = 3.169$. Hence, while you can reject his hypothesis at the 5% significance level, you cannot reject it at the 1% level.

e) Your manager is still somewhat skeptical. Show him/her that the confidence interval approach that you also learned about in your class gives rise to the same conclusion.

Specifically, you recall that the hypothesis can be tested by constructing the confidence interval for $\hat{\beta}_y$ and checking whether the hypothesized value of $\beta_y^* = 0.8$ falls inside this interval or not.

<u>Solution:</u> The confidence interval is given by $(\hat{\beta}_y - t_{crit} \cdot SE(\hat{\beta}_y); \hat{\beta}_y + t_{crit} \cdot SE(\hat{\beta}_y))$.

You remember that for a 95% confidence interval the critical value is $t_{crit} = t_{T-2;5\%}$. Hence, it is the same as for a $t-$test with significance level 5%. You can therefore use the same critical values as before.

With these pieces of information your 95% confidence interval for $\hat{\beta}_y$ is $(0.5-2.228 \cdot 0.125; 0.5+2.228 \cdot 0.125) = (0.2215;\ 0.7785)$. Since your manager's hypothesized true value $\beta_y^* = 0.8$ falls outside this interval, you can again reject his hypothesis based on a 95% confidence interval. You explain to him that this means that if you had observed a 100 different samples of portfolio returns, only in 5 cases would the hypothesized value of $\beta_y^* = 0.8$ fall into that interval.

That said, you also tell him that his hypothesis falls into the 99% confidence interval since $(\hat{\beta}_y - t_{10;1\%} \cdot SE(\hat{\beta}_y); \hat{\beta}_y + t_{10;1\%} \cdot SE(\hat{\beta}_y)) = (0.5-3.17 \cdot 0.125; 0.5+3.17 \cdot 0.125) = (0.104; 0.896)$ includes his hypothesized value of 0.8.

f) Your manager is finally convinced by your elaborate reply, and now asks you to check whether the fund's portfolio has a statistically significant positive alpha. For this exercise, assume that the estimated standard error for the OLS estimate of $\hat{\alpha}_y = 7\%$ was $SE(\hat{\alpha}_y) = 3.5\%$. What null hypothesis do you have to test and against which alternative to answer that question? Conduct the test using the test of significance approach.

Solution: Since your manager asks whether alpha is positive, you perform a one-sided test with null hypothesis $H_0 : \hat{\alpha}_y \leq 0$ vs. the alternative $H_1 : \hat{\alpha}_y > 0$.

You compute the corresponding t-test statistic as

$$t_\alpha = \frac{\hat{\alpha}_y - \alpha^*}{SE(\hat{\alpha}_y)} = \frac{7-0}{3.5} = 2$$

Since this is a one-tailed test, the 5% critical value is now $t_{crit} = t_{10;5\%} = 1.812$. The rejection region for this one-tailed test is thus all values of the test statistic that are greater or equal than 1.812. Hence, since your test statistic is $t_\alpha = 2$, you reject the null that $\alpha_y = 0$ vs the alternative that $\alpha_y > 0$.

You check your result by computing the corresponding confidence interval. This is given by $(\hat{\alpha}_y - t_{10;5\%} \cdot SE(\hat{\alpha}_y); \infty) = (7\% - 1.812 \cdot 3.5\%; \infty) = (0.66; \infty)$. Since $\alpha_y^* = 0$ is not included in the confidence interval you again reject the null hypothesis that your firm's portfolio had an alpha of zero with respect to the benchmark portfolio.